

TARTU ÜLIKOOL
Faculty of Science and Technology
Institute of Computer Science

Hugo Tristan Tammik, Markkus Pokrovski, Thomas Tennisson

KAGGLE - BLACKJACK OUTCOME ANALYSIS

Group project

Repository link: <https://github.com/tennisso/Blackjack-predictor>

Tartu 2025

Business understanding

1. Identifying Business Goals

1.1 Background

Blackjack is one of the most widely played card games in casinos, combining different aspects together such as: probability, decision-making, and financial risk. Players often rely on intuition, incomplete knowledge of odds to guide their betting and playing strategy. However, the availability of large-scale datasets makes it possible to discover patterns in game outcomes and evaluate predictive models. For this project, we use a Kaggle dataset of 900,000 recorded blackjack hands, containing information about cards, sums and round outcomes.

This project is not designed to benefit a commercial business. Instead, it benefits individual learners of data science by demonstrating how data mining techniques can uncover insights in decision-based environments. The project also serves educational purposes within our IDS2025 course, as we, the project authors can implement what we have learned in the course.

1.2 Business Goals

Our business goals are to understand which factors most influence the likelihood of a player winning or losing a blackjack hand. Discover structural patterns, such as card combinations, dealer upcards, and player totals, that correlate with winning outcomes. Build predictive models that estimate the probability of winning based on pre-decision features. Lastly, provide insights how data-driven understanding can improve strategic thinking in uncertain environments.

1.3 Business Success Criteria

Produce a clear, evidence based explanation of the most influential variables affecting blackjack outcomes. Provide visualizations and summaries so anyone can understand the findings even non-technical people. Show that predictive models perform better than basic guesses. Make sure the results are understandable and practical, not just unexplained model outputs.

2. Assessing Situation

2.1 Inventory of Resources:

- Kaggle dataset: 900,000 blackjack hands with detailed card-level information (player cards (up to 5), dealer cards (up to 5), sums, blackjack indicator, win/loss, bust types).
- Tools & Technologies: Python, Jupyter Notebook, pandas, NumPy, visualization libraries.
- Team Skills: data cleaning, feature engineering, model training, evaluation, visualization, and reporting.
- Time & Scope: A course project roughly 2-3 weeks in duration.

2.2 Requirements, Assumptions, and Constraints

- The project must be reproducible and documented according to course expectations.
- We assume all hands in the dataset follow standard blackjack rules, but the dataset may contain variations (such as dealer hit/stand rules).
- Ethical constraints: we are not designing gambling systems or decision-automation tools, the analysis is purely educational.

2.3 Risks and Contingencies

Data quality risk: Inconsistent card encodings or incorrect “sumofcards” values, can be checked by recalculating totals. Model validity risk: Predictions may rely on information not available before decisions (data leakage, avoided by using only pre-decision features). Interpretation risk: High predictive accuracy may give an impression that blackjack can be reliably predicted, which is addressed by contextualizing results within the game’s inherent randomness.

2.4 Terminology

- Blackjack: A hand total of 21 with the first two cards.
- Bust: A total exceeding 21.
- Dealer upcard: The dealer’s visible first card.
- Win/loss: Round outcome for the player.
- Pre-decision features: Information known before the player makes strategic choices (card1, card2, dealer upcard).

2.5 Costs and Benefits

Main costs are time spent cleaning, modeling, and documenting the data and computing effort. The benefits include gaining a deeper, data-driven understanding of blackjack mechanics and clearer insight into how the game behaves.

3. Defining Your Data-Mining Goals

3.1 Data-Mining Goals

The main data-mining goals for this project are to build predictive and descriptive models that help explain which factors influence the outcome of a blackjack hand. Specifically, we aim to develop classification models that estimate the probability of winning, losing based on pre-decision features such as the player's starting hand and the dealer's upcard. Also we want to identify meaningful patterns and relationships within the dataset, such as how different player totals or dealer upcards correlate with outcomes. Furthermore, compare several modelling approaches (logistic regression, decision tree, random forest) to determine which method provides the best balance between predictive accuracy and interpretability. And finally generate insights that help explain why certain starting conditions lead to better or worse outcomes for the player.

3.2 Data-Mining Success Criteria

The data-mining phase will be considered successful if the following criteria are met. Predictive performance: The models should perform better than guessing the most common outcome and achieve reasonable accuracy and ROC-AUC. Interpretability: The most influential features can be clearly identified, and the models provide understandable explanations of why certain predictions are made. Pattern validity: The patterns discovered in the data are logically consistent with blackjack rules and known probability structures. No data leakage: Only pre-decision variables are used in the models to ensure that predictions reflect information actually available at decision time. Actionable insights: The final results offer meaningful explanations of which factors affect blackjack outcomes and help illustrate strategic thinking under uncertainty.

Data understanding

1. Gathering Data

1.1 Data Requirements

For this project, we need a dataset that contains detailed and structured information about Blackjack hands. The most important requirements include information about the player's cards or hand value, the dealer's visible card and final total, the size of the bet, and the final outcome of the hand. These fields allow us to study what affects winning or losing and to build predictive models. Additionally, the data must be large enough to support reliable statistical analysis and machine learning training.

1.2 Verify Data Availability

The dataset used in this project is a public Kaggle dataset containing approximately 900,000 recorded Blackjack hands. It is freely accessible, well-documented, and available as a clean CSV file. Before starting with any analysis, we checked that the file loads correctly, the columns are intact, and the data types are consistent. The dataset size (61.77 MB) is large enough for our analysis and includes a wide variety of game states.

1.3 Define Selection Criteria

Not all columns in the dataset are necessary for achieving the project goals. We focus on variables that directly influence gameplay decisions or contribute to predicting the final outcome. These include player total, dealer upcard, bet amount, and win/loss/push results. Variables unrelated to the decision or outcome - such as internal identifiers or metadata, will be removed. If any columns contain high levels of missing or inconsistent values, they may either be removed or cleaned during the next data preparation step.

2. Describing Data

The dataset contains about 900,000 rows, with each row representing one full Blackjack hand. Each record usually includes fields such as the player's final hand value, the dealer's upcard, the dealer's final total, the action chosen by the player (hit, stand, double, split), the amount bet on the hand, and the resulting outcome. Although the dataset contains player actions and dealer final totals, these are not used in our predictive modeling because they occur after the initial decision state.

Numerical variables such as hand totals, bet sizes, and profits allow for statistical analysis, while categorical variables such as actions and outcomes help identify patterns.

When we look at the simple summaries of the data, we can already see some clear trends. Player totals commonly fall around ordinary Blackjack values such as 15-20. Dealer upcards appear evenly spread across values. Bet sizes vary a lot, suggesting different risk-taking behavior across hands. Outcome frequencies indicate that losing is slightly more common than winning, which matches expectations for casino Blackjack. Overall, the dataset is rich and well-structured for deeper analysis.

3. Exploring Data

Exploring the data involves examining distributions, identifying relationships between variables, and generating early hypotheses. Histograms of player totals reveal typical gameplay patterns, while bar charts of outcome frequencies show the baseline win/loss balance. Scatterplots between bet size and profit help illustrate how much returns can change. Cross-tabulations of dealer upcards versus outcomes help identify situations that are more favorable or risky for the player.

This step also highlights opportunities for feature engineering. For example, identifying soft hands (hands containing an ace counted as 11), calculating the difference between player and dealer visible totals, or counting the number of hits taken could all provide useful features for predictive modeling. Early exploration confirms that the dataset is diverse enough to support both descriptive analysis and predictive modeling.

4. Verifying Data Quality

We checked the data for completeness, consistency, accuracy, and the presence of noise. Most critical columns - such as player total, dealer upcard, and outcome - are fully populated. Logical checks show that hand totals stay within reasonable limits and that outcomes match the rules of Blackjack. The dataset does not appear to have any structural errors, duplicated rows, or corrupted fields. Outliers, such as unusually large bet values, exist but are reasonable within gambling contexts and may require normalization rather than removal.

In general, the dataset is high quality and ready for analysis. Any small issues will be handled in the data preparation phase.

Planning your project

To successfully complete our blackjack outcome analysis, we made a plan to reach our goals and thoroughly work on each step. It contains multiple stages, the estimated time cost and each member's individual working hours. The total expected workload is approximately 31 hours per person.

1. Project Tasks and Time Allocation

1.1 Data Preparation (12 hours total)

Tasks: Load the dataset, check data quality, handle missing values, clean inconsistent records, and prepare final usable fields.

Contribution: Markkus 4h, Hugo 4h, Thomas 4h.

1.2 Exploratory Data Analysis (18 hours total)

Tasks: Compute descriptive statistics, visualize distributions, analyze correlations, and identify gameplay patterns.

Contribution: Markkus 6h, Hugo 6h, Thomas 6h.

1.3 Feature Engineering (12 hours total)

Tasks: Create new features such as soft-hand indicators, advantage metrics and risk-level variables.

Contribution: Markkus 4h, Hugo 4h, Thomas 4h.

1.4 Model Development & Tuning (24 hours total)

Tasks: Build and evaluate logistic regression, decision tree, random forest, and other models; tune hyperparameters; ensure no data leakage.

Contribution: Markkus 8h, Hugo 8h, Thomas 8h.

1.5 Interpretation & Insights (9 hours total)

Tasks: Analyze feature importance, interpret model behavior, and derive strategic conclusions.

Contribution: Markkus 3h, Hugo 3h, Thomas 3h.

1.6 Making the Poster & Presentation (18 hours total)

Tasks: Compile CRISP-DM sections, refine visualizations, and make the poster to present our project, also think, what to speak about on the presentation day.

Contribution: Markkus 6h, Hugo 6h, Thomas 6h.

2. Tools and Methods

We will use Python, pandas, NumPy, scikit-learn, matplotlib/seaborn, Jupyter Notebook, and GitHub. Methodologically, the project follows CRISP-DM, incorporating EDA, feature engineering, classification modelling, evaluation metrics, and interpretability analysis.

2.1 Important Notes

Only pre-decision features will be used to avoid data leakage.

All work will be committed to the GitHub repository for reproducibility.

Visualizations will prioritize clarity and pedagogical value over complexity.